Sabuj Laskar

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LinkedIn | GitHub | Google Scholar | Website

Research Interests

Computer Architecture, AI Accelerator Architecture, LLM Inference, 3D-DRAM-Based Architectures, Network-on-Chip and Chiplet Interconnects, Performance Modeling, Collective Communication

Publications

1. **Sabuj Laskar**, Hyojong Kim, Devesh Singh, Mrinmoy Ghosh, and Eun Jung Kim, "SYMPHONY: Harmonizing Memory and Communication in LLM Inference on 3D DRAM Accelerators," in Proceedings of the 59th IEEE/ACM International Symposium on Microarchitecture (**MICRO**), 2026.

2. Pranati Majhi, **Sabuj Laskar**, Abdullah Muzahid and Eun Jung Kim, "Compression-Aware Gradient Splitting for Collective Communications in Distributed Training," 2026 IEEE International Symposium on High-Performance Computer Architecture (**HPCA**), 2026.

3. **Sabuj Laskar**, Pranati Majhi, Abdullah Muzahid and Eun Jung Kim, "SuperMesh: Energy-Efficient Collective Communications for Accelerators," In 58th IEEE/ACM International Symposium on Microarchitecture (**MICRO**), 2025 (**Distinguished Artifact Award**).

4. **Sabuj Laskar**, Pranati Majhi, Sungkeun Kim, Farabi Mahmud, Abdullah Muzahid and Eun Jung Kim, "Enhancing Collective Communication in MCM Accelerators for Deep Learning Training," 2024 IEEE International Symposium on High-Performance Computer Architecture (**HPCA**), 2024.

Education

Texas A&M University — College Station, Texas
Ph.D. in Computer Science; Advisor: *Dr. Eun Jung Kim* — 08/22–12/27 (Expected)

Bangladesh University of Engineering and Technology — Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering — 02/15–04/19

Work Experience

Samsung Semiconductor, Inc. — San Jose, California
Computer Architecture Research Intern, AGI Computing Lab — 05/25–01/26

- Characterized memory and communication bottlenecks across six latency-dominant operators in LLM decoding on 3D-DRAM-based accelerators.
- Developed Symphony, an operator-aware mapping framework, and BEAM, a collective communication algorithm, achieving 1.35–1.67× decode-latency speedup over existing weight-partitioning strategies (MICRO 2026).

Texas A&M University — College Station, Texas
Graduate Research Assistant, High Performance Computing Lab — 08/22–Present

Research Experience

Adaptive All-to-All (A2A) for Skewed Communication in MoE Inference — 05/25–Present
Supervisors: *Dr. Eun Jung Kim*, *Dr. Abdullah Muzahid*

- Identified prohibitive schedule-generation overheads for global A2A optimization under dynamic and skewed MoE communication patterns.
- Observed stable per-expert load and designed hot-expert-aware adaptive routing using local information.
- Achieved up to 1.66× speedup over SOTA A2A schedulers on skewed MoE workloads.

Energy-Efficient Collective Communications for Accelerators 12/23–06/25

Supervisors: *Dr. Eun Jung Kim*, *Dr. Abdullah Muzahid*

- Designed an enhanced mesh-based accelerator interconnect topology with additional peripheral links to reduce collective communication latency.
- Evaluated architectural performance and hardware area/power overheads using Synopsys and DSENT.
- Achieved 1.19–1.75× speedup over conventional mesh-based architectures.

Collective Communication Algorithms for Multi-Chip Modules (MCMs) 08/22–11/23

Supervisors: *Dr. Eun Jung Kim*, *Dr. Abdullah Muzahid*

- Designed topology-aware AllReduce algorithms, including tree-based TTO and bidirectional Ring, for mesh-connected multi-chip AI accelerators.
- Achieved over 30% reduction in training time compared with state-of-the-art collective algorithms.

Skills

- **Programming:** C/C++, Python
- **Architecture & Performance Modeling:** BookSim, Astra-Sim, ScaleSim, LLMCompass
- **Hardware & Analysis Tools:** Synopsys, DSENT
- **Parallel & ML Systems:** PyTorch, NCCL, OpenMPI, SLURM, Docker, Singularity

Professional Service

- **Program Committee Member:** MICRO 2026
- **Artifact Evaluation Committee Member:** HPCA 2026
- **Reviewer:** IEEE CAL 2026

Awards and Honors

- Student Travel Award, MICRO 2026
- Distinguished Artifact Award, MICRO 2025
- Student Travel Award, MICRO 2025
- Student Travel Award, HPCA 2024